

Proof. Degenerate cases. If $a = 0$ or $b = 0$ then $ab = 0$ while $\frac{a^p}{p} + \frac{b^q}{q} \geq 0$; the inequality holds with equality. Hence we assume $a > 0$ and $b > 0$ for the remainder of the proof.

Weighted AM–GM inequality. Let $\lambda \in [0, 1]$ and $u, v \geq 0$. Since $\log t$ is concave on $(0, \infty)$, Jensen's inequality gives

$$\lambda \log u + (1 - \lambda) \log v \leq \log(\lambda u + (1 - \lambda)v).$$

Exponentiating (the exponential function is monotone increasing) yields

$$u^\lambda v^{1-\lambda} \leq \lambda u + (1 - \lambda)v. \quad (2)$$

Equality in (2) holds exactly when $u = v$ (or when $\lambda \in \{0, 1\}$, which cannot occur here because $p, q > 1$).

Choice of the weight. Because $\frac{1}{p} + \frac{1}{q} = 1$ with $p, q > 1$, the numbers

$$\lambda := \frac{1}{p}, \quad 1 - \lambda := \frac{1}{q}$$

satisfy $\lambda, 1 - \lambda \in (0, 1)$ and $\lambda + (1 - \lambda) = 1$.

Apply (2) to $u = a^p$ and $v = b^q$. Substituting $u = a^p$, $v = b^q$ and the above λ into (2) we obtain

$$(a^p)^{1/p} (b^q)^{1/q} \leq \frac{1}{p} a^p + \frac{1}{q} b^q.$$

Simplify the left-hand side. Since $(a^p)^{1/p} = a$ and $(b^q)^{1/q} = b$,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q},$$

which is the desired inequality.

Equality condition. Equality in the weighted AM–GM step (2) occurs precisely when $a^p = b^q$. When $a = b = 0$ both sides of Young's inequality are zero and the condition $a^p = b^q = 0$ is satisfied. Consequently, equality holds *iff* $a^p = b^q$ (including the trivial zero case). $\square$